Chapter 28

Campbell Biology, AP Edition, 12e

Image of a spiraling, squash shaped organism that is many times larger than a group of six organisms adjacent. All appear to be aquatic, single-celled organisms. Scale bar: 1 micrometer.

Chapter 28********

Protists

Figure 28.1

Knowing that prokaryotes are extremely small, you might assume that this image shows six prokaryotes and one much larger eukaryote. But in fact, the only prokaryote is the organism immediately above the scale bar —the rest are members of diverse, mostly unicellular groups of eukaryotes informally known as protists.

A diagram of endosymbiosis among eukaryotes.

A schematic representation of the process of endosymbiosis attributing to protistan diversity is shown. The title is What gave rise to the great diversity of protists, and how have their lineages diverged over time? Much of protistan diversity can be traced to endosymbiosis. An ancestral cell has engulfed an aerobic bacterium and forms an early eukaryote (a heterotroph) with the engulfed aerobic bacterium labeled a mitochondrion. This eukaryotic cell engulfs a photosynthetic bacterium (in green) to form an early photosynthetic eukaryote. The engulfed photosynthetic bacterium is labeled a plastid.

A small box shows the highly diverse subgroups that eukaryotes are divided into: Excavata, SAR, Archaeplastida, and Unikonta, shown similar to a branching phylogenetic tree.

Key Concepts

- 28.1**Most eukaryotes are single-celled organisms**
- 28.2**Excavates include protists with modified mitochondria and protists with unique flagella**
- 28.3**SAR is a highly diverse group of protists defined by DNA similarities**
- 28.4**Red algae and green algae are the closest relatives of plants**

- 28.5**Unikonts include protists that are closely related to fungi and animals**
- 28.6**Protists play key roles in ecological communities**

Advanced placement edition

Protists represent a very diverse group of organisms. Structural evidence such as membrane-bound organelles **(BIG IDEAS 2 & 4)** and research techniques in biotechnology **(BIG IDEA 3)** support the relatedness of all eukaryotes **(BIG IDEA 1)**.

Study Tip

Label a drawing: Protist life cycles vary greatly. To make sense of this variation, it can be helpful to identify when mitosis occurs in different life cycles, as shown here for a portion of Figure 28.14. Identify the stages at which mitosis occurs in Figures 28.14, 28.18, 28.25, and 28.28.

The places where mitosis occurs are marked along a partial life cycle diagram.

The protist life cycle is depicted as a small diagram, with one section of it enlarged to show the steps from zoospores to the formation of female gametophytes. The mitotic steps in the cycle are to be labeled.

Get Ready for This Chapter

In this chapter, you will learn about the origin and evolutionary history of protists, as well as their impact on ecosystems and human societies. Before beginning this chapter, you should be able to do the following:

- Explain the concept of domains (see Concept 26.6, Figure 26.21).
- Describe the endosymbiotic origin of mitochondria and plastids (see Concept 25.3, Figure 25.10).
- Compare the outcomes of mitosis and meiosis (see Figure 13.10).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following domains consists of all the organisms whose cells have true nuclei?

```
Animalia
```

That's incorrect. Try again.

Incorrect. Review the following material: The domain Bacteria includes most of the currently known prokaryotes, while the domain Archaea consists of a diverse group of prokaryotic organisms that inhabit a

wide variety of environments. The domain Eukarya includes many groups of single-celled organisms as well as plants, fungi, and animals.

```
Bacteria
```

That's incorrect. Try again.

Incorrect. Review the following material: The domain Bacteria includes most of the currently known prokaryotes, while the domain Archaea consists of a diverse group of prokaryotic organisms that inhabit a wide variety of environments. The domain Eukarya includes many groups of single-celled organisms as well as plants, fungi, and animals.

```
Archaea
```

That's incorrect. Try again.

Incorrect. Review the following material: The domain Bacteria includes most of the currently known prokaryotes, while the domain Archaea consists of a diverse group of prokaryotic organisms that inhabit a wide variety of environments. The domain Eukarya includes many groups of single-celled organisms as well as plants, fungi, and animals.

```
Eukarya
```

That's correct.

Correct! If you had trouble with this question, review the following material: The three domains—Bacteria, Archaea, and Eukarya—are a taxonomic level higher than the kingdom level. The domain Bacteria includes most of the currently known prokaryotes, while the domain Archaea consists of a diverse group of prokaryotic organisms that inhabit a wide variety of environments. The domain Eukarya consists of all the organisms that have cells containing true nuclei, including many groups of single-celled organisms as well as plants, fungi, and animals.

2. According to the hypothesis of serial endosymbiosis, which of the following was a key initial step that best describes the origin of chloroplasts?

```
A prokaryotic host cell engulfed an aerobic,
photosynthetic bacterium.
```

That's incorrect. Try again.

Incorrect. Review the following material: Evidence indicates that the eukaryotes originated by endosymbiosis when a prokaryotic cell engulfed a small aerobic, heterotrophic bacterium whose descendants evolved into an organelle found in all eukaryotes, the mitochondrion. The prokaryotic ancestor of the mitochondrion probably entered the host cell as undigested prey or an internal parasite. Later, a eukaryotic host cell engulfed a photosynthetic bacterium, leading to the evolution of plastids.

```
        A prokaryotic host cell engulfed an aerobic,
heterotrophic bacterium.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Evidence indicates that the eukaryotes originated by endosymbiosis when a prokaryotic cell engulfed a small aerobic, heterotrophic bacterium whose descendants evolved into an organelle found in all eukaryotes, the mitochondrion. The prokaryotic ancestor of the mitochondrion probably entered the host cell as undigested prey or an internal parasite. Later, a eukaryotic host cell engulfed a photosynthetic bacterium, leading to the evolution of plastids.

```
        A eukaryotic host cell engulfed an aerobic,
photosynthetic bacterium. (new correct answer)
```

That's incorrect. Try again.

Incorrect. Review the following material: Evidence indicates that the eukaryotes originated by endosymbiosis when a prokaryotic cell engulfed a small aerobic, heterotrophic bacterium whose descendants evolved into an organelle found in all eukaryotes, the mitochondrion. The prokaryotic ancestor of the mitochondrion probably entered the host cell as undigested prey or an internal parasite. Later, a eukaryotic host cell engulfed a photosynthetic bacterium, leading to the evolution of plastids.

```
        A eukaryotic host cell engulfed an aerobic,
heterotrophic bacterium.
```

That's incorrect. Try again.

Incorrect. Review the following material: Evidence indicates that the eukaryotes originated by endosymbiosis when a prokaryotic cell engulfed a small aerobic, heterotrophic bacterium whose descendants

evolved into an organelle found in all eukaryotes, the mitochondrion. The prokaryotic ancestor of the mitochondrion probably entered the host cell as undigested prey or an internal parasite. Later, a eukaryotic host cell engulfed a photosynthetic bacterium, leading to the evolution of plastids.

3. A diploid (2n) cell divides by meiosis, producing four daughter cells. Next, each daughter cell divides by mitosis, producing a total of eight granddaughter cells. Which of the following statements about the granddaughter cells is true?

```
All of the granddaughter cells are genetically identical.
```

That's incorrect. Try again.

Incorrect. Review the following material: In meiosis, a parent cell produces four daughter cells, each of which has half the chromosome number of the parent cell. In mitosis, a parent cell produces two daughter cells, each of which is genetically identical to the parent cell.

Each granddaughter cell is diploid (2n).

That's incorrect. Try again.

Incorrect. Review the following material: In meiosis, a parent cell produces four daughter cells, each of which has half the chromosome number of the parent cell. In mitosis, a parent cell produces two daughter cells, each of which is genetically identical to the parent cell.

Each granddaughter cell is haploid (n).

That's correct.

Correct! If you had trouble with this question, review the following material: In meiosis, a parent cell produces four daughter cells, each of which has half the chromosome number of the parent cell; in this case, the diploid (2n) parent cell produces four daughter cells, each of which is haploid (n). In mitosis, a parent cell produces two daughter cells that are genetically identical to the parent cell; in this case, the four haploid daughter cells produce a total of eight granddaughter cells, each of which is haploid (n).

```
Four of the granddaughter cells are genetically identical.
```

That's incorrect. Try again.

Incorrect. Review the following material: In meiosis, a parent cell produces four daughter cells, each of which has half the chromosome number of the parent cell. In mitosis, a parent cell produces two daughter cells, each of which is genetically identical to the parent cell.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 28.1Describe the process of secondary endosymbiosis and explain its role in eukaryotic history.
- 28.2Characterize the excavates.
- 28.3Give examples of the protists classified in SAR.
- 28.4Describe characteristics of red and green algae.
- 28.5Identify and describe the closest eukaryotic relatives of fungi and animals.
- 28.6Give examples of ecological roles played by protists.

Concept 28.1: Most eukaryotes are single-celled organisms

Protists—along with plants, animals, and fungi—are eukaryotes; they are in domain Eukarya, one of the three domains of life. Genetic and morphological studies have shown that some protists are more closely related to plants, fungi, or animals than they are to other protists. As a result, the kingdom in which all protists once were classified, Protista, has been abandoned, and various protist lineages are now recognized as major groups in their own right. Most biologists still use the term *protist*, but only as a convenient way to refer to eukaryotes that are not plants, animals, or fungi.

Unlike the cells of prokaryotes, the cells of protists and other eukaryotes have a nucleus and other membrane-enclosed organelles, such as mitochondria and the Golgi apparatus. Such organelles provide specific locations where particular cellular functions are accomplished, making the structure and organization of eukaryotic cells more complex than those of prokaryotic cells. Eukaryotic cells also have a well-developed cytoskeleton that extends throughout the cell (see Figure 6.20). The cytoskeleton provides the structural support that enables eukaryotic cells to have asymmetric (irregular) forms, as well as to change in shape as they feed, move, or grow. In contrast, prokaryotic cells lack a well-developed cytoskeleton, thus limiting the extent to which they can maintain asymmetric forms or change shape over time.

We'll survey the diversity of eukaryotes throughout the rest of this unit, beginning in this chapter with the protists. As you explore this material, bear in mind that

- the organisms in most eukaryotic lineages are protists, and
- most protists are unicellular.

Thus, life differs greatly from how most of us commonly think of it. The large, multicellular organisms that we know best (plants, animals, and fungi) are the tips of just a few branches on the great tree of life (see Figure 26.21).

HHMI BioInteractive Video

Seeing the Invisible: Van Leeuwenhoek's First Glimpses of the Microbial World

image

image

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

Structural and Functional Diversity in Protists

Given that they are classified in a number of different groups, it isn't surprising that few general characteristics of protists can be cited without exceptions. In fact, protists exhibit more structural and functional diversity than the eukaryotes with which we are most familiar—plants, animals, and fungi.

For example, most protists are unicellular, although there are some colonial and multicellular species. Single-celled protists are justifiably considered the simplest eukaryotes, but at the cellular level, many protists are very complex—the most elaborate of all cells. In multicellular organisms, essential biological functions are carried out by organs. Unicellular protists carry out the same essential functions, but they do so using subcellular organelles, not multicellular organs. The organelles that protists use are mostly those discussed in Figure 6.8, including the nucleus, endoplasmic reticulum, Golgi apparatus, and lysosomes. Certain protists also rely on organelles not found in most other eukaryotic cells, such as contractile vacuoles that pump excess water from the protistan cell (see Figure 7.13). One group of dinoflagellate protists has the most complex subcellular structure known: a unique, eye-like organelle called an *ocelloid* with components resembling the lens and retina in the eyes of vertebrates and some other animals **(Figure 28.2)**.

Figure 28.2

The eye-like ocelloid of *Erythropsidinium*, a dinoflagellate.

A micrograph shows the eye like ocelloid of Erythropsidinium, a dinoflagellate, depicting complex protist cellular structure.

This complex, subcellular structure resembles the camera-like eye found in vertebrates and some molluscs. In some dinoflagellates, the ocelloid may function in prey detection.

A micrograph shows a subcellular structure resembling a lateral view of an eye-like structure, termed ocelloid. It has an irregular outline with distinct layers from right to left. The first layer is followed by an elliptical structure labeled the lens, followed by circular and curved layers. A thick, curved elongated structure is present labeled the retina and seems to be embedded in underlying subcellular structures.

Protists are also very diverse in their nutrition. Some protists are photoautotrophs and contain chloroplasts. Some are heterotrophs, absorbing organic molecules or ingesting larger food particles. Still other protists, called **mixotrophs**, combine photosynthesis and heterotrophic nutrition. Photoautotrophy, heterotrophy, and mixotrophy have all arisen independently in many different protist lineages.

Reproduction and life cycles also are highly varied among protists. Some protists are only known to reproduce asexually; others can also reproduce sexually or at least employ the sexual processes of meiosis and fertilization. All three basic types of sexual life cycles (see Figure 13.6) are represented among protists, along with some variations that do not quite fit any of these types. We will examine the life cycles of several protist groups later in this chapter.

Endosymbiosis in Eukaryotic Evolution

What gave rise to the enormous diversity of protists that exist today? There is abundant evidence that much of protistan diversity has its origins in **endosymbiosis**, a relationship between two species in which one organism lives inside the cell or cells of another organism (the host). In particular, as we discussed in Concept 25.3, structural, biochemical, and DNA sequence data indicate that mitochondria and plastids are derived from bacteria that were engulfed by the ancestors of early eukaryotic cells. The evidence also suggests that mitochondria evolved before plastids. Thus, a defining moment in the origin of eukaryotes occurred when a host cell engulfed a bacterium that would later become an organelle found in all eukaryotes —the mitochondrion.

To determine which bacterial lineage gave rise to mitochondria, researchers have compared the DNA sequences of mitochondrial genes (mtDNA) to those found in major clades of bacteria. In the **Scientific Skills Exercise**, you will interpret one such set of DNA sequence comparisons. Collectively, such studies indicate that mitochondria arose from an alpha proteobacterium (see Figure 27.17). Results from mtDNA sequence analyses also indicate that the mitochondria of protists, animals, fungi, and plants descended from a single common ancestor, thus suggesting that mitochondria arose only once over the course of evolution.

Similar analyses provide evidence that plastids descended from a single common ancestor—a cyanobacterium that was engulfed by a eukaryotic host cell.

Scientific Skills Exercise

Interpreting Comparisons of Genetic Sequences

AP Science Practices 2, 3, 4

Which Bacteria Are Most Closely Related to Mitochondria? Early eukaryotes acquired mitochondria by endosymbiosis: A host cell engulfed an aerobic bacterium that persisted within the cytoplasm to the mutual benefit of both cells. In studying which living bacteria might be most closely related to mitochondria, researchers compared DNA sequences encoding ribosomal RNA (rRNA), a component of ribosomes. Ribosomes perform critical cell functions. Hence, genes encoding rRNA are under strong selection and change slowly over time, making them suitable for comparing even distantly related species. In this exercise, you will interpret some of the research data to draw conclusions about the phylogeny of mitochondria.

A field of wheat.

Wheat, used as the source of mitochondrial rRNA

How the Research Was Done Researchers isolated and cloned DNA sequences from the gene that codes for the small-subunit rRNA molecule for the wheat (a eukaryote) mitochondrion and for five bacterial species:

- Wheat, used as the source of genes encoding mitochondrial rRNA
- *Agrobacterium tumefaciens*, an alpha proteobacterium that lives within plant tissue and produces tumors in the host
- *Comamonas testosteroni*, a beta proteobacterium
- *Escherichia coli*, a well-studied gamma proteobacterium that inhabits human intestines
- *Mycoplasma capricolum*, a gram-positive mycoplasma, which is the only group of bacteria lacking cell walls
- *Anacystis nidulans*, a cyanobacterium

Data from the Research Gene sequences encoding rRNA for the wheat mitochondrion and the five bacteria were aligned and compared. The data table below, called a *comparison matrix*, summarizes the comparison of 617 nucleotide positions from the gene sequences. Each value in the table is the percentage of

the 617 nucleotide positions for which the pair of organisms have the same base. Any positions that were identical across all six rRNA-encoding gene sequences were omitted from this comparison matrix.

A table illustrates the similarity matrix score for wheat mitochondrion, A. tumefaciens, C. testosteroni, E. coli, M. capricolum, and A. nidulans.

	Wheat mitochondrion	**A. tumefaciens**	**C. testosteroni**	**E. coli**	**M. capricolum**
Wheat mitochondrion	-	48	38	35	34
A. tumefaciens		-	55	57	52
C. testosteroni			-	61	52
E. coli				-	48
M. capricolum					-
A. nidulans					

Data from D. Yang et al., Mitochondrial origins, *Proceedings of the National Academy of Sciences USA* 82:4443–4447 (1985).

INTERPRET THE DATA

- First, make sure you understand how to read the comparison matrix.
- Find the cell that represents the comparison of *C. testosteroni* and *E. coli*. What value is given in this cell?
- What does that value signify about the comparable gene sequences in those two organisms?
- Explain why some cells have a dash rather than a value.
- Why are some cells shaded gray, with no value?
- Why did the researchers choose one plant mitochondrion and five bacterial species to include in the comparison matrix?
-
  - Which bacterium has an rRNA-encoding gene that is most similar to that of the wheat mitochondrion?

- What is the significance of this similarity?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Progress has also been made toward identifying the host cell that engulfed an alpha proteobacterium, thereby setting the stage for the origin of eukaryotes. For example, researchers recently reported the discovery of a new group of archaea, the lokiarchaeotes. In phylogenomic analyses, this group was identified as the sister group of the eukaryotes and its genome was found to encode many eukaryote-specific features. Was the host cell that engulfed an alpha proteobacterium a lokiarchaeote? While this could be correct, it is also possible that the host belonged to a different lineage of archaea or was closely related to the archaea (but was not itself an archaean). In any case, current evidence indicates that the host was a relatively complex cell in which certain features of eukaryotic cells had evolved, such as a cytoskeleton that enabled it to change shape (and thereby engulf the alpha proteobacterium).

Plastid Evolution: A Closer Look

As you've seen, current evidence indicates that mitochondria are descended from a bacterium that was engulfed by a host cell that was an archaean (or a close relative of the archaea). This event gave rise to the eukaryotes. There is also much evidence that later in eukaryotic history, a lineage of heterotrophic eukaryotes acquired an additional endosymbiont—a photosynthetic cyanobacterium—that then evolved into plastids. According to the hypothesis illustrated in **Figure 28.3**, this plastid-bearing lineage gave rise to two lineages of photosynthetic protists, or ********algae**: red algae and green algae.

Figure 28.3

```
Diversity of plastids produced by endosymbiosis.
```

Through different pathways, primary and secondary endosymbiosis result in stramenopiles, alveolates, euglenids and chlorarachniophytes.

Studies of plastid-bearing eukaryotes suggest that plastids evolved from a cyanobacterium that was engulfed by an ancestral heterotrophic eukaryote (primary endosymbiosis). That ancestor then diversified into red algae and green algae, some of which were subsequently engulfed by other eukaryotes (secondary endosymbiosis).

Heterotrophic eukaryote (large cell with nucleus) engulfs a cyanobacterium (much smaller cell with no nucleus) in primary endosymbiosis. The cyanobacterium inside the eukaryote has three membranes (labeled 1, 2 and 3); one of which will be lost in red and green algal descendants. Two pathways follow:

(1) Cell goes to a red alga, which undergoes secondary endosymbiosis, and is engulfed by another large eukaryotic cell. That cell goes to (a) Stramenopiles with plastid and nucleus and (b) Alveolates with plastid, nucleus and flagellum.

(2) Cell goes to a green alga, and two additional pathways diverge. (2a) Green alga undergoes secondary endosymbiosis and is engulfed by a larger eukaryotic cell, resulting in the Euglenids with plastid, nucleus and flagella. (2b) Green alga undergoes secondary endosymbiosis and is engulfed by a larger eukaryotic cell, resulting in Chlorarachniophytes with plastid and nucleus and pseudopodia.

VISUAL SKILLS *Based on this diagram, which of the following groups are likely to be more closely related: stramenopiles and alveolates, or euglenids and chlorarachniophytes? Explain.*

*For suggested answer, see**Appendix A.*

Let's examine some of the steps in Figure 28.3 more closely. First, recall that cyanobacteria are gram-negative and that gram-negative bacteria have two cell membranes, an inner plasma membrane and an outer membrane that is part of the cell wall (see Figure 27.3). Plastids in red algae and green algae are also surrounded by two membranes. Transport proteins in these membranes are homologous to proteins in the inner and outer membranes of cyanobacteria, providing further support for the hypothesis that plastids originated from a cyanobacterial endosymbiont.

On several occasions during eukaryotic evolution, red algae and green algae underwent **secondary endosymbiosis**, meaning they were ingested in the food vacuoles of heterotrophic eukaryotes and became endosymbionts themselves. For example, protists known as chlorarachniophytes likely evolved when a heterotrophic eukaryote engulfed a green alga. Evidence for this process can be found within the engulfed cell, which contains a tiny vestigial nucleus, called a *nucleomorph***(Figure 28.4)**. Genes from the nucleomorph are still transcribed, and their DNA sequences indicate that the engulfed cell was a green alga.

Figure 28.4

```
Nucleomorph within a plastid of a chlorarachniophyte.
```

Image of a nucleomorph within a plastid of a chlorarachniophyte. Nucleomorph is surrounded by inner and outer plastid membrane and nuclear pore-like gap is present.

Four Supergroups of Eukaryotes

Our understanding of the evolutionary history of eukaryotic diversity has been in flux in recent years. Not only has kingdom Protista been abandoned, but other hypotheses have been discarded as well. For example,

many biologists once thought that the first lineage to have diverged from all other eukaryotes was the *amitochondriate protists*, organisms without conventional mitochondria and with fewer membrane-enclosed organelles than other protist groups. But recent structural and DNA data have undermined this hypothesis. Many of the so-called amitochondriate protists have been shown to have mitochondria—though reduced ones—and some of these organisms are now classified in distantly related groups.

The ongoing changes in our understanding of the phylogeny of protists pose challenges to students and instructors alike. Hypotheses about these relationships are a focus of scientific activity, changing rapidly as new data cause previous ideas to be modified or discarded. We'll focus here on one current hypothesis: the four supergroups of eukaryotes shown in **Figure 28.5**. Because the root of the eukaryotic tree is not known, all four supergroups are shown as diverging simultaneously from a common ancestor. We know that this is not correct, but we do not know which supergroup was the first to diverge from the other three. In addition, major new groups of protists have been discovered in recent years, but their relationship to the supergroups shown in Figure 28.5 has yet to be resolved. Examples of such unresolved groups include haptophytes and cryptophytes **(Figure 28.6)**, as well as the hemimastigophores, a novel group described in 2018. Overall, as you read this chapter, it may be helpful to focus less on the specific names of groups of organisms and more on why the organisms are important and how ongoing research is elucidating their evolutionary relationships.

Figure 28.5

```
Exploring Protistan Diversity
```

A phylogenetic tree of protistan diversity. Examples and descriptions for three of the supergroups.

The following tree represents a phylogenetic hypothesis for the relationships among eukaryotes on Earth today. The eukaryotic groups at the branch tips are related in larger "supergroups," labeled vertically at the far right of the tree. Groups that were formerly classified in the kingdom Protista are highlighted in yellow. Dotted lines indicate evolutionary relationships that are uncertain. For clarity, this tree only includes representative clades from each supergroup. In addition, the recent discoveries of many new groups of eukaryotes indicate that eukaryotic diversity is actually much greater than shown here. The diagram is as follows.

Branch point 1

Super Group Excavata (Branch point 2 a)

Branch point 3 a

Diplomonads (Protista)

Parabasalids (Protista)

Euglenozoans (Protista)

Super Group “SAR” clade (Branch point 2 b, dotted line)

Branch point 3 b 1 (dotted line)

(Stramenopiles dotted line) Branch point 4 b 1

Diatoms (Protista)

Golden algae (Protista)

Brown algae (Protista)

(Alveolates, dotted line) Branch point 4 b 2

Branch point 5 b 1

Dinoflagellates (Protista)

Apicomplexans (Protista)

Ciliates (Protista)

(Rhizarians, dotted line) Branch point 3 b 2

Branch point 4 b 3

Radiolarians (Protista)

Forams (Protista)

Cercozoans (Protista)

Super Group Archaeplastida (Branch point 2 c)

Branch point 3 c

(Green algae) Chlorophytes (Protista)

Branch point 4c

(Green algae) Charophytes (Protista)

Plants

Red algae (Protista)

Super Group Unikonta (Branch point 2 d, dotted line)

(Amoebozoans, dotted line) Branch point 3 d 1

Slime molds (Protista)

Tubulinids (Protista)

Entamoebas (Protista)

(Opisthokonts, dotted line) Branch point 3 d 2

Branch point 4 d 1

Nucleariids (Protista)

Fungi

Branch point 4 d 2

Choanoflagellates (Protista)

Animals

Notes on Excavata are as follows. A micrograph shows Giardia, which is roughly tear-shaped single cell with several attached string-like structures (flagella). Scale: 5 micrometers. Excavata: Some members of this supergroup have an “excavated” groove on one side of the cell body. Two major clades (the parabasalids and diplomonads) have highly reduced mitochondria; members of a third clade (the euglenozoans) have flagella that differ in structure from those of other organisms. Excavates include parasites such as Giardia, as well as many predatory and photosynthetic species. Photo: Giardia intestinalis, a diplomonad parasite. This diplomonad (colorized SEM), which lacks the characteristic surface groove of the Excavata, inhabits the intestines of mammals. It can infect people when they drink water contaminated with feces containing Giardia cysts. Drinking such water, even from a seemingly pristine stream, can cause severe diarrhea. Boiling the water kills the parasite.

Parts are as follows.

SAR clade. Image: Diatom diversity and Globigerina, a rhizarian in the SAR clade. This supergroup contains (and is named after) three large and very diverse clades: Stramenopila, Alveolata, and Rhizaria. Stramenopiles include some of the most important photosynthetic organisms on Earth, such as the diatoms shown here. Alveolates also include photosynthetic species, as well as important pathogens, such as Plasmodium, which causes malaria. According to one current hypothesis, stramenopiles and alveolates

originated by secondary endosymbiosis when a heterotrophic protist engulfed a red alga. Photo. Diatom diversity. These beautiful single-celled protists are important photosynthetic organisms in aquatic communities (LM). The rhizarian subgroup of SAR includes many species of amoebas, most of which have pseudopodia that are threadlike in shape. Pseudopodia are extensions that can bulge from any portion of the cell; they are used in movement and in the capture of prey. Photo. Globigerina, a rhizarian in SAR. This species is a foram, a group whose members have threadlike pseudopodia that extend through pores in the shell, or test (LM). The inset shows a foram test, which is hardened by calcium carbonate.

Archaeplastida. This supergroup of eukaryotes includes red algae and green algae, along with plants. Red algae and green algae include unicellular species, colonial species, and multicellular species (including the green alga Volvox). Many of the large algae known informally as “seaweeds” are multicellular red or green algae. Protists in Archaeplastida include key photosynthetic species that form the base of the food web in many aquatic communities. Photo. Volvox, a multicellular freshwater green alga. This alga has two types of differentiated cells, and so it is considered multicellular rather than colonial. It resembles a hollow ball whose wall is composed of hundreds of biflagellated cells (see inset LM) embedded in a gelatinous extracellular matrix; if isolated, these cells cannot reproduce. However, the alga also contains cells that are specialized for either sexual or asexual reproduction. The large algae shown here will eventually release the small “daughter” algae that can be seen within them (LM).

Unikonta. This supergroup of eukaryotes includes amoebas that have lobe- or tube-shaped pseudopodia, as well as animals, fungi, and non-amoeba protists that are closely related to animals or fungi. According to one current hypothesis, the unikonts were the first eukaryotic supergroup to diverge from all other eukaryotes; however, this hypothesis has yet to be widely accepted. Photo. A unikont amoeba. This amoeba, the tubulinid Amoeba proteus, is using its pseudopodia to move.

DRAW IT *Draw a simplified version of this phylogenetic tree that shows only the four supergroups of eukaryotes. Now sketch how the tree would look if the unikonts were the sister group to all other eukaryotes.*

*For suggested answer, see**Appendix A.*

Video

Volvox

image

Play

image

Video

Amoeba Pseudopodium

image

Play

image

Figure 28.6

```
A haptophyte and a cryptophyte.
```

Part a. A micrograph of a spherical haptophyte. Part b. A micrograph of cryptophytes with flagella.

Although they are major groups of protists, haptophytes and cryptophytes are not shown in Figure 28.5 because their phylogenetic relationships have yet to be resolved. ***(a)*** *The haptophyte Emiliania huxleyi is a key producer in the ocean.* ***(b)*** *The outer surface of this photosynthetic cryptophyte is covered with protective plates. (SEM)*

Part a. A micrograph of a spherical haptophyte, Emiliania huxleyi. It resembles a globe-shape with many overlapping subunits. Each subunit resembles two circles, attached by radiating lines (similar to a wheel). Scale bar 2 micrometers.

Part b. A micrograph of cryptophytes with flagella. An image of an elliptical organism with a tail-like flagellum is shown. The oval part is covered by scales. Scale bar: 8 micrometer.

Concept Check 28.1

- Cite at least four examples of structural and functional diversity among protists.
- Summarize the role of endosymbiosis in eukaryotic evolution.
- **MAKE CONNECTIONS** After studying Figure 28.3, predict how many distinct genomes are contained within the cell of a chlorarachniophyte. Explain. (See Figures 6.17 and 6.18.)

*For suggested answers, see*Appendix A.**

Concept 28.2: Excavates include protists with modified mitochondria and protists with unique flagella

A phylogenetic tree highlights excavate, which includes diplomondas, parabasalids, and euglenozoans.

Now that we have examined some of the broad patterns in eukaryotic evolution, we will look more closely at the four main groups of protists shown in Figure 28.5.

We begin with **Excavata** (the excavates), a clade that was originally proposed based on morphological studies of the cytoskeleton. Some members of this diverse group also have an "excavated" feeding groove on one side of the cell body. The excavates include the diplomonads, parabasalids, and euglenozoans. Molecular data indicate that each of these three groups is monophyletic. However, most recent phylogenomic studies do not support the monophyly of the excavates as a whole, making Excavata the most controversial of our four supergroups.

Diplomonads and Parabasalids

The protists in these two groups lack plastids and have highly reduced mitochondria (until recently, they were thought to lack mitochondria altogether). Most diplomonads and parabasalids are found in anaerobic environments.

Diplomonads have reduced mitochondria called *mitosomes*. These organelles lack functional electron transport chains and hence cannot use oxygen to help extract energy from carbohydrates and other organic molecules. Instead, diplomonads get the energy they need from anaerobic biochemical pathways. Many diplomonads are parasites, including the infamous *Giardia intestinalis* (see Figure 28.5), which inhabits the intestines of mammals.

Structurally, diplomonads have two equal-sized nuclei and multiple flagella. Recall that eukaryotic flagella are extensions of the cytoplasm, consisting of bundles of microtubules covered by the cell's plasma membrane (see Figure 6.24). They are quite different from prokaryotic flagella, which are filaments composed of globular proteins attached to the cell surface (see Figure 27.7).

Parabasalids also have reduced mitochondria; called *hydrogenosomes*, these organelles generate some energy anaerobically, releasing hydrogen gas as a by-product. The best-known parabasalid is *Trichomonas vaginalis*, a sexually transmitted parasite that infects about 140 million people each year worldwide. *T. vaginalis* travels along the mucus-coated lining of the human reproductive and urinary tracts by moving its

flagella and by undulating part of its plasma membrane **(Figure 28.7)**. In females, if the vagina's normal acidity is disturbed, *T. vaginalis* can outcompete beneficial microorganisms there and infect the vagina. One of *T. vaginalis*'s genes encodes a product that allows it to feed on the vaginal lining, promoting infection. Studies suggest that *T. vaginalis* acquired this gene by horizontal gene transfer from bacterial parasites in the vagina. (*Trichomonas* infections also can occur in the urethra of males, though often without symptoms.)

Figure 28.7

```
The parabasalid parasite *Trichomonas vaginalis.*
```

Parabasalid parasite with undulating membrane and flagella. Scale: 5 micrometers.

(colorized SEM)

Euglenozoans

Protists called **euglenozoans** belong to a diverse clade that includes predatory heterotrophs, photosynthetic autotrophs, mixotrophs, and parasites. The main morphological feature that distinguishes protists in this clade is the presence of a rod with either a spiral or a crystalline structure inside each of their flagella **(Figure 28.8)**. The two best-studied groups of euglenozoans are the kinetoplastids and the euglenids.

Figure 28.8

```
Euglenozoan flagellum.
```

Diagram of a euglenozoan with flagella (scale: 8 micrometers) with cross section (TEM, scale 0.2 micrometers) of a flagellum with "stripes" of crystalline rod and ring of microtubules visible.

Most euglenozoans have a crystalline rod inside one of their flagella (the TEM is a flagellum shown in cross section). The rod lies alongside the 9+2 ring of microtubules found in all eukaryotic flagella (compare with Figure 6.24).

Kinetoplastids

Protists called **kinetoplastids** have a single, large mitochondrion that contains an organized mass of DNA called a *kinetoplast*. These protists include species that feed on prokaryotes in freshwater, marine, and moist terrestrial ecosystems, as well as species that parasitize animals, plants, and other protists. For example,

kinetoplastids in the genus *Trypanosoma* infect humans and cause sleeping sickness **(Figure 28.9)**, a neurological disease that is invariably fatal if not treated. This disease currently afflicts about 10,000 people each year, mostly in rural areas of Africa. The infection occurs via the bite of a vector (carrier) organism, the African tsetse fly. Trypanosomes also cause Chagas' disease, which is transmitted by bloodsucking insects and can lead to congestive heart failure.

Figure 28.9

```
    *Trypanosoma*, the kinetoplastid that causes sleeping
sickness.
```

Trypanosoma are ribbon-like cells that appear about twice as long as the disk-shaped red blood cells in this image. Scale: 9 micrometers.

(colorized SEM)

Trypanosomes evade immune responses with an effective "bait-and-switch" defense. The surface of a trypanosome is coated with millions of copies of a single protein. However, before the host's immune system can recognize the protein and mount an attack, new generations of the parasite switch to another surface protein with a different molecular structure. Frequent changes in the surface protein prevent the host from developing immunity. (The Scientific Skills Exercise in Chapter 43 explores this topic further.) About a third of *Trypanosoma*'s genome is dedicated to producing these surface proteins.

Euglenids

A **euglenid** has a pocket at one end of the cell from which one or two flagella emerge **(Figure 28.10)**. Some euglenids are mixotrophs: They perform photosynthesis when sunlight is available, but when it is not, they can become heterotrophic, absorbing organic nutrients from their environment. Many other euglenids engulf prey by phagocytosis.

Figure 28.10

```
    *Euglena*, a euglenid commonly found in pond water.
```

Structures and their functions in Euglena image (scale: 5 micrometers) and diagram. Euglena is an elongated tear-shaped organism with long flagellum, light detector, pellicle and eyespot.

The long flagellum is a whip-like structure extending from one end of the cell. The eyespot (on the same end of the cell as the flagellum) is a pigmented organelle that functions as a light shield, allowing light from only

a certain direction to strike the light detector, which is a swelling near the base of the long flagellum. It detects light that is not blocked by the eyespot. As a result, Euglena moves toward light of appropriate intensity, an important adaptation that enhances photosynthesis. The pellicle is protein bands beneath the plasma membrane that provide strength and flexibility (Euglena lacks a cell wall). Many chloroplasts are visible as round discs inside the cell, as is the large nucleus and contractile vacuole, and the short flagellum (at the base of the long flagellum).

Video

Euglena

image

Play

image

Play

Concept Check 28.2

- Why do some biologists describe the mitochondria of diplomonads and parabasalids as "highly reduced"?
- **WHAT IF?** DNA sequence data for a diplomonad, a euglenid, a plant, and an unidentified protist suggest that the unidentified species is most closely related to the diplomonad. Further studies reveal that the unknown species has fully functional mitochondria. Based on these data, at what point on the phylogenetic tree in Figure 28.5 did the mystery protist's lineage probably diverge from other eukaryote lineages? Explain.

*For suggested answers, see*Appendix A.**

Concept 28.3: SAR is a highly diverse group of protists defined by DNA similarities

A phylogenetic tree highlights the SAR group.

Our second supergroup, referred to as **SAR**, was proposed recently based on whole-genome DNA sequence analyses. These studies found that three major clades of protists—the stramenopiles, alveolates, and rhizarians—form a monophyletic supergroup. This supergroup contains a large, extremely diverse collection

of protists. To date, this supergroup has not received a formal name but is instead known by the first letters of its major clades: SAR.

Some morphological and DNA sequence data suggest that two of these groups, the stramenopiles and alveolates, originated more than a billion years ago, when a common ancestor of these two clades engulfed a single-celled, photosynthetic red alga. Because red algae are thought to have originated by primary endosymbiosis (see Figure 28.3), such an origin for the stramenopiles and alveolates is referred to as secondary endosymbiosis. Others question this idea, noting that some species in these groups lack plastids or their remnants (including any trace of plastid genes in their nuclear DNA).

Despite its lack of a formal name, the SAR supergroup represents the best current hypothesis for the phylogeny of the three large protist clades to which we now turn.

Stramenopiles

One major subgroup of SAR, the **stramenopiles**, includes some of the most important photosynthetic organisms on the planet. Their name (from the Latin *stramen*, straw, and *pilos*, hair) refers to their characteristic flagellum, which has numerous fine, hairlike projections. In most stramenopiles, this "hairy" flagellum is paired with a shorter "smooth" (nonhairy) flagellum **(Figure 28.11)**. Here we'll focus on three groups of stramenopiles: diatoms, oomycetes, and brown algae.

Figure 28.11

```
Stramenopile flagella.
```

Stramenopile have hairy and smooth flagella. Scale: 5 micrometers.

Most stramenopiles, such as Synura petersenii, have two flagella: one covered with fine, stiff hairs and a shorter one that is smooth.

Diatoms

A key group of photosynthetic protists, **diatoms** are unicellular algae that have a unique glass-like wall made of silicon dioxide embedded in an organic matrix **(Figure 28.12)**. The wall consists of two parts that overlap like a shoe box and its lid. These walls provide effective protection from the crushing jaws of predators: Live diatoms can withstand pressures as great as 1.4 million kg/m2, equal to the pressure under each leg of a table supporting an elephant!

Figure 28.12

```
The diatom *Triceratium morlandii.*
```

Diatom pictured is roughly triangular in shape with rounded “points” of the triangle and has a tough-looking outer edge. Scale: 40 micrometers.

(colorized SEM)

With an estimated 100,000 living species, diatoms are a highly diverse group of protists (see Figure 28.5). They are among the most abundant photosynthetic organisms both in the ocean and in lakes: One bucket of water scooped from the surface of the sea may contain millions of these microscopic algae. The abundance of diatoms in the past is also evident in the fossil record, where massive accumulations of fossilized diatom walls are major constituents of sediments known as *diatomaceous earth*. These sediments are mined for their quality as a filtering medium and for many other uses.

Diatoms are so widespread and abundant that their photosynthetic activity affects global carbon dioxide (CO_2) levels. Diatoms have this effect in part because of events that occur during episodes of rapid population growth, or *blooms*, when ample nutrients are available. Typically, diatoms are eaten by a variety of protists and invertebrates, but during a bloom, many escape this fate. When these uneaten diatoms die, their bodies sink to the ocean floor. It takes decades, or even centuries, for dead diatoms that sink to the ocean floor to be broken down by bacteria and other decomposers. As a result, the carbon in their bodies remains there for some time, rather than being released immediately as CO_2 as the decomposers respire. The overall effect of these events is that CO_2 absorbed by diatoms during photosynthesis is transported, or “pumped,” to the ocean floor.

With an eye toward reducing global warming by lowering atmospheric CO_2 levels, some scientists advocate promoting diatom blooms by fertilizing the ocean with essential nutrients such as iron. In one study testing this idea, researchers found that CO_2 was indeed pumped to the ocean floor after iron was added to a small region of the ocean, causing a diatom bloom. Other scientists have cautioned that the small area and short duration of such studies makes it difficult to predict what would happen if large-scale iron fertilization experiments were performed. Indeed, the possibility that iron fertilization could have undesirable side effects (such as oxygen depletion or the production of nitrous oxide, a more potent greenhouse gas than CO_2) led the 2008 United Nations Convention on Biological Diversity to issue a moratorium against such experiments.

Brown Algae

The largest and most complex algae are **brown algae**. All are multicellular, and most are marine. Brown algae are especially common along temperate coasts that have cold-water currents. They owe their characteristic brown or olive color to the carotenoids in their plastids.

Many of the species commonly called “seaweeds” are brown algae. Some brown algal seaweeds have specialized structures that resemble organs in plants, such as a rootlike **holdfast**, which anchors the alga, and a stemlike **stipe**, which supports the leaflike **blades(Figure 28.13)**. Unlike plants, however, brown algae lack true tissues and organs. Moreover, morphological and DNA data show that these similarities evolved independently in the algal and plant lineages and are thus analogous, not homologous. In addition, while plants have adaptations (such as rigid stems) that provide support against gravity, brown algae have adaptations that enable their main photosynthetic surfaces (the leaflike blades) to be near the water surface. Some brown algae accomplish this task with gas-filled, bubble-shaped floats. Giant brown algae known as kelps that live in deep waters have such floats in their blades, which are attached to stipes that can rise as much as 60 m from the seafloor—more than half the length of a football field.

Figure 28.13

```
Seaweeds: adapted to life at the ocean’s margins.
```

Sea palm attached to rocky bottom by holdfast (resembles a branched plant root). Stipe is stalk-like coming out of holdfast and blades, at top, are flat and leaf-like in appearance.

The sea palm (Postelsia) lives on rocks along the coast of the northwestern United States and western Canada. The body of this brown alga is well adapted to maintaining a firm foothold despite the crashing surf.

Brown algae are important commodities for humans. Some species are eaten, such as *Laminaria* (Japanese “kombu”), which is used in soups. In addition, the cell walls of brown algae contain a gel-forming substance, called algin, which is used to thicken many processed foods, including pudding and salad dressing.

Alternation of Generations

A variety of life cycles have evolved among the multicellular algae. The most complex life cycles include an ****alternation of generations**, the alternation of multicellular haploid and diploid forms, as shown in **Figure 28.14** for the brown alga *Laminaria*. Although haploid and diploid conditions alternate in *all* sexual life cycles—human gametes, for example, are haploid—the term *alternation of generations* applies only to life cycles in which both haploid and diploid stages are multicellular. As you will read in Concept 29.1, alternation of generations also evolved in plants.

Figure 28.14

The life cycle of the brown alga *Laminaria:***an example of alternation of generations.**

Diagram outlines alternation of generations in Laminaria, from the diploid sporophyte through the haploid gametophyte generation and back to the sporophyte.

1. The sporophytes (2n) are usually found in water just below the line of the lowest tides, attached to the rocks by branching holdfast, with stipe and blades extending up towards the surface.
2. Cells on the surface of the blade develop into sporangia.
3. Sporangia produce haploid zoospores with both hairy and smooth flagella by meiosis.
4. The zoospores are all structurally alike, but about half of them develop into male gametophytes and half into female gametophytes. (Both are haploid). The gametophytes are short, branched filaments that grow on subtidal rocks.
5. Male gametophytes release haploid sperm, and female gametophytes produce haploid eggs, which remain attached to the female gametophyte. Eggs secrete a chemical signal that attracts sperm of the same species, thereby increasing the probability of fertilization in the ocean.
6. Sperm fertilize the eggs on the female gametophyte, resulting in diploid zygotes still on the haploid mature female gametophytes.
7. The zygotes grow into new diploid multicellular sporophytes while attached to the remains of the haploid female gametophyte. The diploid sporophyte continues to develop until the cycle begins again (#1). Photograph of a cluster of sporophyte blades is shown at scale 10 centimeters.

VISUAL SKILLS *Based on this diagram, are the sperm shown in* **STEP 5** *genetically identical to one another? Explain.*

*For suggested answer, see**Appendix A.*

As detailed in Figure 28.14, the diploid individual in this complex algal life cycle is called the *sporophyte* because it produces spores. The spores are haploid and move by means of flagella; they are called zoospores. The zoospores divide by mitosis and develop into haploid, multicellular male and female *gametophytes*, which produce gametes. The union of two gametes (fertilization) results in a diploid zygote, which matures and gives rise to a new multicellular sporophyte.

In *Laminaria*, the two generations are **heteromorphic**, meaning that the sporophytes and gametophytes are structurally different. Other algal life cycles have an alternation of **isomorphic** generations, in which the sporophytes and gametophytes look similar to each other, although they differ in chromosome number.

Oomycetes (Water Molds and Their Relatives)

Oomycetes include the water molds, the white rusts, and the downy mildews. Based on their morphology, these organisms were previously classified as fungi (in fact, oomycete means “egg fungus”). For example, many oomycetes have multinucleate filaments (hyphae) that resemble fungal hyphae **(Figure 28.15a)**. However, there are key differences between oomycetes and fungi. Among them, oomycetes typically have cell walls made of cellulose, whereas the walls of fungi consist mainly of another polysaccharide, chitin. Data from molecular systematics have confirmed that oomycetes are not closely related to fungi. In both oomycetes and fungi, the high surface-to-volume ratio of filamentous structures enhances the uptake of nutrients from the environment.

Figure 28.15

```
Oomycetes.
```

Two micrographs. Part a. Closeup of comycete hyphae. Part b. Water mold growing on a goldfish.

Although oomycetes descended from plastid-bearing ancestors, they no longer have plastids and do not perform photosynthesis. Instead, they typically acquire nutrients as decomposers or parasites. Most water molds are decomposers that grow as cottony masses on dead algae and animals, mainly in freshwater habitats **(Figure 28.15b)**. White rusts and downy mildews generally live on land as plant parasites.

The ecological impact of oomycetes can be significant. For example, the oomycete *Phytophthora infestans* causes potato late blight, which turns the stalks and stems of potato (and tomato) plants into black slime. Late blight contributed to the devastating Irish famine of the 19th century, in which a million people died and at least that many were forced to leave Ireland. The disease remains a major problem today, causing crop losses as high as 70% in some regions. Globally, about $1 billion is spent each year on pesticides to control the parasite.

Alveolates

Members of the next subgroup of SAR, the **alveolates**, have membrane-enclosed sacs (alveoli) just under the plasma membrane **(Figure 28.16)**. Alveolates are abundant in many habitats and include a wide range of photosynthetic and heterotrophic protists. We’ll discuss three alveolate clades here: a group of flagellates (the dinoflagellates), a group of parasites (the apicomplexans), and a group of protists that move using cilia (the ciliates).

Figure 28.16

```
Alveoli.
```

A micrograph and diagram show alveoli sacs adjacent to the flagellum.

These sacs under the plasma membrane are a characteristic that distinguishes alveolates from other eukaryotes (TEM).

Dinoflagellates

The cells of many **dinoflagellates** are reinforced by cellulose plates. Grooves in this “armor” house two flagella, one of which causes dinoflagellates (from the Greek *dinos*, whirling) to spin as they move through the waters of their marine and freshwater communities **(Figure 28.17a)**.

Figure 28.17

```
Dinoflagellates.
```

Two photos of a dinoflagellate and a red tide.

Parts are as follows.

Part a. Image: Spherical cell with groove in center (resembles a hamburger); spiral flagella inside groove. Scale: 3 micrometers. Dinoflagellate flagella. Beating of the spiral flagellum, which lies in a groove that encircles the cell, makes this specimen of Pfiesteria shumwayae spin (colorized SEM).

Part b. Image: Beach with red water lapping against sand and bluer water further out. Red tide in the Gulf of Carpentaria in northern Australia. The red color is due to high concentrations of a carotenoid-containing dinoflagellate.

Video

Dinoflagellate

image

Play

image

Play

Although their ancestors may have originated by secondary endosymbiosis (see Figure 28.3), roughly half of all dinoflagellate species are now purely heterotrophic. Others are important species of *phytoplankton* (photosynthetic plankton, which include photosynthetic bacteria as well as algae); many photosynthetic dinoflagellates are mixotrophic.

Periods of explosive population growth (blooms) in dinoflagellates sometimes cause a phenomenon called "red tide" **(Figure 28.17b)**. The blooms make coastal waters appear brownish red or pink because of the presence of carotenoids, the most common pigments in dinoflagellate plastids. When blooms occur, toxins produced by certain dinoflagellates have caused massive kills of invertebrates and fishes. Humans who eat molluscs that have accumulated the toxins are affected as well, sometimes fatally. Based on data from 1982–2016, researchers at Stony Brook University found that ocean warming (caused by ongoing climate change) has expanded the potential for algal blooms in two species of toxic dinoflagellates, making them a growing threat to human health.

Apicomplexans

Nearly all **apicomplexans** are parasites of animals—and virtually all animal species examined so far are attacked by these parasites. The parasites spread through their host as tiny infectious cells called *sporozoites*. Apicomplexans are so named because one end (the *apex*) of the sporozoite cell contains a *complex* of organelles specialized for penetrating host cells and tissues. Although apicomplexans are not photosynthetic, recent data show that they retain a modified plastid (apicoplast), most likely of red algal origin.

Most apicomplexans have intricate life cycles with both sexual and asexual stages. Those life cycles often require two or more host species for completion. For example, *Plasmodium*, the parasite that causes malaria, lives in both mosquitoes and humans **(Figure 28.18)**.

Figure 28.18

```
    The two-host life cycle of *Plasmodium*, the apicomplexan
that causes malaria.
```

Person is bitten by infected Anopheles mosquito; Plasmodium sporozoites enter the liver cells, and then merozoites infect red blood cells. Gametocytes are produced and lifecycle continues in another mosquito host.

Diagram depicts Plasmodium life cycle in both the human and the mosquito host.

1.An infected Anopheles mosquito bites a person, injecting Plasmodium sporozoites (haploid) in its saliva.

2.The sporozoites enter the person's liver cells. After several days, the sporozoites undergo multiple divisions and become merozoites (haploid), which use their apical complex to penetrate red blood cells. Diagram shows numerous merozoites spilling out of a liver cell and penetrating red blood cells. TEM shows a merozoite on the surface of a red blood cell attaching at its apex. Scale: 0.5 micrometers.

3.The merozoites divide asexually inside the red blood cells. At intervals of 48 or 72 hours (depending on the species), large numbers of merozoites break out of the blood cells, causing periodic chills and fever. Some of the merozoites infect other red blood cells.

4.Some merozoites form gametocytes (haploid).

5.Another Anopheles mosquito bites the infected person and picks up Plasmodium gametocytes along with blood.

6.Gametes (haploid) form from gametocytes; each male gametocyte produces several slender male gametes. Female gametes are also formed.

7.Fertilization occurs in the mosquito's digestive tract, and a diploid zygote forms.

8.Following meiosis, an oocyst (haploid) develops from the zygote in the wall of the mosquito's gut. The oocyst releases thousands of haploid sporozoites, which migrate to the mosquito's salivary gland. Cycle can repeat at step (1).

Are morphological differences between sporozoites, merozoites, and gametocytes caused by different genomes or by differences in gene expression? Explain.

*For suggested answer, see**Appendix A.*

Animation

Life Cycle of a Malaria Parasite

image

image

Historically, malaria has rivaled tuberculosis as the leading cause of human death by infectious disease. The incidence of malaria was diminished in the 1960s by insecticides that reduced carrier populations of *Anopheles* mosquitoes and by drugs that killed *Plasmodium* in humans. But the emergence of resistant varieties of both *Anopheles* and *Plasmodium* has led to a resurgence of malaria. According to the World Health Organization, there were an estimated 241 million malaria cases and 627,000 malaria deaths worldwide in 2020. In regions where malaria is common, the lethal effects of this disease have resulted in

the evolution of high frequencies of the sickle-cell allele; for an explanation of this connection, see Figure 23.18.

The search for malarial vaccines has been hampered by the fact that *Plasmodium* lives mainly inside cells, hidden from the host's immune system. And, like trypanosomes, *Plasmodium* continually changes its surface proteins. Even so, significant progress was made when European regulators recently approved the world's first licensed malarial vaccine (RTS,S or Mosquirix). By 2019, selected regions of Africa began routine use of this vaccine, and in 2021 the World Health Organization recommended it for broad use in children in areas where malaria is prevalent. However, this vaccine, which targets a protein on the surface of sporozoites, provides only partial protection against malaria. As a result, researchers continue to study other potential vaccine targets, including the apicoplast. This approach may be effective because the apicoplast is a modified plastid; as such, it descended from a cyanobacterium and hence has different metabolic pathways from those in human patients.

Ciliates

The **ciliates** are a large and varied group of protists named for their use of cilia to move and feed **(Figure 28.19a)**. Most ciliates are predators, typically of bacteria or of other protists. Their cilia may completely cover the cell surface or may be clustered in a few rows or tufts. In certain species, rows of tightly packed cilia function collectively in locomotion. Other ciliates scurry about on leg-like structures constructed from many cilia bonded together.

Figure 28.19

```
    Structure and function in the ciliate *Paramecium caudatum.*
```

Two diagrams. Part a. Feeding, waste removal and water balance. Part b. Conjugation and reproduction.

Parts are as follows.

Part a. Paramecium constantly takes in water by osmosis from its hypotonic environment. Bladderlike contractile vacuoles accumulate excess water from radial canals and periodically expel it through the plasma membrane. Thousands of cilia cover the surface of the Paramecium. Cilia along a funnel-shaped oral groove move food (mainly bacteria) into the cell mouth, where the food is engulfed into food vacuoles by phagocytosis. Food vacuoles fuse with lysosomes (not shown). As the food is digested, the vacuoles follow a looping path through the cell. Wastes are released when the vacuoles fuse with a specialized region of the plasma membrane that functions as an anal pore. In the image, micronucleus are adjacent to and smaller than macronucleus. Contractile vacuoles (quantity 2) are distinguishable as star-like structures; food vacuoles as spheres; cilia as short "hairs" on cell surface. Scale: 50 micrometers.

Part b. Conjugation (Steps 1-4):

1. Two cells of compatible mating strains align side by side and partially fuse. The micronuclei are diploid.
2. Following meiosis, four haploid micronuclei are produced in each cell.
3. Three micronuclei in each cell disintegrate. The remaining micronucleus in each cell divides by mitosis.
4. The cells swap one micronucleus.

Asexual reproduction (Steps 5-9):

5. The cells separate.
6. Following micronuclear fusion, the two haploid micronuclei in the cell fuse into a diploid micronucleus.
7. Three rounds of mitosis produce 8 micronuclei inside the cell.
8. Four micronuclei become macronuclei and the original macronucleus disintegrates.
9. Two rounds of binary fission yield four daughter cells. Cells that are compatible mates can now continue at step 1.

MAKE CONNECTIONS *The events shown in steps* **4** *through* **6** *of this diagram have a similar overall effect to what event in the human life cycle (see*Figure 13.5*)*? Explain.*

*For suggested answer, see**Appendix A*.

Video

Paramecium

image

Play

image

Play

A distinctive feature of ciliates is the presence of two types of nuclei: tiny micronuclei and large macronuclei. A cell has one or more nuclei of each type. Genetic variation results from ****conjugation, a sexual process in which two individuals exchange haploid micronuclei but do not reproduce **(Figure 28.19b)**. Ciliates generally reproduce asexually by binary fission, during which the existing macronucleus disintegrates and a new one is formed from the cell's micronuclei. Each macronucleus typically contains multiple copies of the ciliate's genome. Genes in the macronucleus control the everyday functions of the cell, such as feeding, waste removal, and maintaining water balance.

Video

Ciliate Movement in *Stentor*

image

Play

image

Rhizarians

Our next subgroup of SAR is the **rhizarians**. Many species in this group are **amoebas**, protists that move and feed by means of ****pseudopodia, extensions that may bulge from almost anywhere on the cell surface. An amoeba moves by extending a pseudopodium and anchoring its tip; more cytoplasm then streams into the pseudopodium. Amoebas do not constitute a monophyletic group; instead, they are dispersed across many distantly related eukaryotic taxa. Most amoebas that are rhizarians differ morphologically from other amoebas by having threadlike pseudopodia. Rhizarians also include flagellated (non-amoeboid) protists that feed using threadlike pseudopodia.

We'll examine three groups of rhizarians here: radiolarians, forams, and cercozoans.

Radiolarians

The protists called **radiolarians** have delicate, intricately symmetrical internal skeletons that are generally made of silica. The pseudopodia of these mostly marine protists radiate from the central body **(Figure 28.20)** and are reinforced by bundles of microtubules. The microtubules are covered by a thin layer of cytoplasm, which engulfs smaller microorganisms that become attached to the pseudopodia. Cytoplasmic streaming then carries the captured prey into the main part of the cell. After radiolarians die, their skeletons settle to the seafloor, where they have accumulated as an ooze that is hundreds of meters thick in some locations.

Figure 28.20

```
A radiolarian.
```

Radiolarian. Scale: 200 micrometers.

Numerous threadlike pseudopodia radiate from the central body of this radiolarian (LM).

Forams

The protists called **foraminiferans** (from the Latin *foramen*, little hole, and *ferre*, to bear), or **forams**, are named for their porous shells, called **tests** (see Figure 28.5). Foram tests consist of a single piece of organic material that typically is hardened with calcium carbonate. The pseudopodia that extend through the pores function in swimming, test formation, and feeding. Many forams also derive nourishment from the photosynthesis of symbiotic algae that live within the tests.

Forams are found in both the ocean and fresh water. Most species live in sand or attach themselves to rocks or algae, but some live as plankton. The largest forams, though single-celled, have tests several centimeters in diameter.

Ninety percent of all identified species of forams are known from fossils. Along with the calcium-containing remains of other protists, the fossilized tests of forams are part of marine sediments, including sedimentary rocks that are now land formations. Foram fossils are excellent markers for correlating the ages of sedimentary rocks in different parts of the world. Researchers are also studying these fossils to obtain information about climate change and its effects on the oceans and their life **(Figure 28.21)**.

Figure 28.21

```
Fossil forams.
```

Fossil foram tests embedded in rock. Forams are roughly tear-shaped and lined with layers of lines.

By measuring the magnesium content in fossilized forams like these, researchers seek to learn how ocean temperatures have changed over time. Forams take up more magnesium in warmer water than in colder water.

Cercozoans

First identified in molecular phylogenies, the **cercozoans** are a large group of amoeboid and flagellated protists that feed using threadlike pseudopodia. Cercozoan protists are common inhabitants of marine, freshwater, and soil ecosystems.

Most cercozoans are heterotrophs. Many are parasites of plants, animals, or other protists; many others are predators. The predators include the most important consumers of bacteria in aquatic and soil ecosystems, along with species that eat other protists, fungi, and even small animals. One small group of cercozoans, the chlorarachniophytes (mentioned earlier in the discussion of secondary endosymbiosis), are mixotrophic: These organisms ingest smaller protists and bacteria as well as perform photosynthesis. At least one other cercozoan, *Paulinella chromatophora*, is an autotroph, deriving its energy from light and its carbon from CO_2. *Paulinella* **(Figure 28.22)** appears to represent an intriguing additional evolutionary example of a eukaryotic lineage that obtained its photosynthetic apparatus directly from a cyanobacterium.

Figure 28.22

```
A second case of primary endosymbiosis?
```

Cercozoan Paulinella with chromatophore highlighted.

The cercozoan Paulinella conducts photosynthesis in a unique sausage-shaped structure called a chromatophore (LM). Chromatophore membranes include a peptidoglycan layer, indicating that they are derived from a bacterium. DNA evidence shows that chromatophores are derived from a different cyanobacterium than that from which plastids are derived.

Concept Check 28.3

- Explain why forams have such a well-preserved fossil record.
- **WHAT IF?** Would you expect the plastid DNA of photosynthetic dinoflagellates and diatoms to be more similar to the nuclear DNA of plants (domain Eukarya) or to the chromosomal DNA of cyanobacteria (domain Bacteria)? Explain.
- **MAKE CONNECTIONS** Which of the three life cycles in Figure 13.6 exhibits alternation of generations? How does it differ from the other two?
- **MAKE CONNECTIONS** Review Figures 9.2 and 10.5, and then summarize how CO_2 and O_2 are both used and produced by chlorarachniophytes and other aerobic algae.

*For suggested answers, see*Appendix A.**

Concept 28.4: Red algae and green algae are the closest relatives of plants

A phylogenetic tree highlights group archeplastida.

As described earlier, morphological and molecular evidence indicates that plastids arose when a heterotrophic protist acquired a cyanobacterial endosymbiont. Later, photosynthetic descendants of this ancient protist evolved into red algae and green algae (see Figure 28.3), and the lineage that produced green algae then gave rise to plants. Together, red algae, green algae, and plants make up our third eukaryotic supergroup, which is called **Archaeplastida**. Archaeplastida is a monophyletic group that descended from the ancient protist that engulfed a cyanobacterium. We will examine plants in Chapters 29 and 30; here we will look at the diversity of their closest algal relatives, red algae and green algae.

Red Algae

Many of the 6,000 known species of **red algae** (rhodophytes, from the Greek *rhodos*, red) are reddish, owing to the photosynthetic pigment phycoerythrin, which masks the green of chlorophyll **(Figure 28.23)**. However, other species (those adapted to shallow water) have less phycoerythrin. As a result, red algal species may be greenish red in very shallow water, bright red at moderate depths, and almost black in deep water. Some species lack pigmentation altogether and live as heteroptrophic parasites on other red algae.

Figure 28.23

```
Red algae.
```

Photos show three types of red algae. Bonnemaisonia hamifera, dulse (Palmaria palmata), and nori (Porphyra), with a photo showing how nori is used to wrap sushi.

Types are as follows.

Bonnemaisonia hamifera. This red alga has a filamentous form.

Dulse (Palmaria palmata). This edible species has a leafy form.

Nori. The red alga Porphyra is the source of a traditional Japanese food. The seaweed is grown on nets in shallow coastal waters. Paper-thin, glossy sheets of dried nori make a mineral-rich wrap for rice, seafood, and vegetables in sushi.

Red algae are abundant in the warm coastal waters of tropical oceans. Some of their photosynthetic pigments, including phycoerythrin, allow them to absorb blue and green light, which penetrate relatively far into the water. A species of red alga has been discovered near the Bahamas at a depth of more than 260 m. There are also a small number of freshwater and terrestrial species.

Most red algae are multicellular. Although none are as big as the giant brown kelps, the largest multicellular red algae are included in the informal designation “seaweeds.” You may have eaten one of these multicellular red algae, *Porphyra* (Japanese “nori”), as crispy sheets or as a wrap for sushi (see Figure 28.23). Red algae reproduce sexually and have diverse life cycles in which alternation of generations is common. However, unlike other algae, red algae do not have flagellated gametes, so they depend on water currents to bring gametes together for fertilization.

Green Algae

The grass-green chloroplasts of **green algae** have a structure and pigment composition much like the chloroplasts of plants. Molecular systematics and cellular morphology leave little doubt that green algae and plants are closely related. In fact, some systematists now advocate including green algae in an expanded “plant” kingdom, Viridiplantae (from the Latin *viridis*, green). Phylogenetically, this change makes sense, since otherwise the green algae are a paraphyletic group.

Green algae can be divided into two main groups, the charophytes and the chlorophytes. The charophytes include the algae most closely related to plants, and we will discuss them along with plants in Chapter 29.

The second group, the chlorophytes (from the Greek *chloros*, green), includes more than 7,000 species. Most live in fresh water, but there are also many marine and some terrestrial species. The simplest chlorophytes are unicellular organisms such as *Chlamydomonas*, which resemble gametes of more complex chlorophytes. Various species of unicellular chlorophytes live independently in aquatic habitats as phytoplankton or inhabit damp soil. Some live symbiotically within other eukaryotes, contributing part of their photosynthetic output to the food supply of their hosts. Still other chlorophytes live in environments exposed to intense visible and ultraviolet radiation; these species are protected by radiation-blocking compounds in their cytoplasm, cell wall, or zygote coat.

Larger size and greater complexity evolved in green algae by three different mechanisms:

- The formation of colonies of individual cells, as seen in *Pediastrum***(Figure 28.24a)** and other species that contribute to the stringy masses known as pond scum.
- The formation of true multicellular bodies by cell division and differentiation, as in *Volvox* (see Figure 28.5) and *Ulva***(Figure 28.24b)**.
- The repeated division of nuclei with no cytoplasmic division, as in *Caulerpa***(Figure 28.24c)**.

Figure 28.24

```
    Examples of large chlorophytes.
```

Three micrographs of large chlorophytes.

Chlorophytes are as follows.

Part a. Pediastrum, a pond alga. Chlorophytes in this genus form daughter colonies with the same number and arrangement of cells as the parent colony.

Part b. Ulva, or sea lettuce. This multicellular, edible chlorophyte has differentiated structures, such as its leaflike blades and a rootlike holdfast that anchors the alga.

Part c. Caulerpa, an intertidal chlorophyte. The branched filaments lack crosswalls and thus are multinucleate. In effect, the body of this alga is one huge supercell.

Most chlorophytes have complex life cycles, with both sexual and asexual reproductive stages. Nearly all species of chlorophytes reproduce sexually by means of biflagellated gametes that have cup-shaped chloroplasts **(Figure 28.25)**. Alternation of generations has evolved in some chlorophytes, including *Ulva*.

Figure 28.25

```
    The life cycle of *Chlamydomonas*, a unicellular
chlorophyte.
```

Life cycle including both haploid and diploid stages, and TEM of Chlamydomonas showing cell with a cross section of cup-shaped chloroplast, nucleus, cell wall, and 2 flagella. Scale: 1 micrometer.

In this life cycle diagram, the processes of meiosis, fertilization, asexual and sexual reproduction, and diploid and haploid stages are highlighted.

1.In Chlamydomonas, mature cells are haploid and contain a single cup-shaped nucleus.

1.Sexual reproduction (Steps 2 to 5):

2.In response to a nutrient shortage, drying of the environment, or other stress, cells develop into gametes (haploid).

3.Gametes of different mating types (designated + and -) fuse (fertilization), forming a diploid zygote.

4.The zygote (2n) secretes a durable coat that protects the cells from harsh conditions.

5.After a dormant period, meiosis produces four haploid individuals (two of each mating type) that emerge and mature.

6.Asexual reproduction (Steps 6 to 7):

7.When a mature cell reproduces asexually, it resorbs its flagella and then undergoes two rounds of mitosis, forming four haploid cells (more in some species).

8.These haploid daughter cells develop flagella and cell walls and then emerge as swimming zoospores (haploid) from the parent cell. The zoospores develop into mature haploid cells and can repeat at step 1.

VISUAL SKILLS *Circle the stage(s) in the diagram in which clones are formed, producing additional new daughter cells that are genetically identical to the parent cell(s).*

*For suggested answer, see**Appendix A.*

Animation

Alternation of Generations in a Protist

image

Play

image

Play

Concept Check 28.4

- Contrast red algae and brown algae.
- Why is it accurate to say that *Ulva* is truly multicellular but *Caulerpa* is not?
- **WHAT IF?** Suggest a possible reason why species in the green algal lineage may have been more likely to colonize land than species in the red algal lineage.

*For suggested answers, see*Appendix A.**

Concept 28.5: Unikonts include protists that are closely related to fungi and animals

A phylogenetic tree highlights group Unikonta.

Unikonta (also called Amorphea) is an extremely diverse supergroup of eukaryotes that includes animals, fungi, and some protists. There are two major clades of unikonts, the amoebozoans (tubulinids and close protist relatives) and the opisthokonts (animals, fungi, and closely related protist groups). Each of these two major clades is strongly supported by molecular systematics. Support for the close relationship between amoebozoans and opisthokonts is provided by comparisons of myosin proteins and by some (but not all) studies based on multiple genes or whole genomes.

An ongoing debate involving the unikonts concerns the root of the eukaryotic tree. Recall that the root of a phylogenetic tree anchors the tree in time: Branch points close to the root are the oldest. At present, the root of the eukaryotic tree is uncertain; hence, we do not know which supergroup of eukaryotes was the first to diverge from all other eukaryotes. If the root of the eukaryotic tree were known, it would help scientists infer characteristics of the common ancestor of all eukaryotes.

In trying to determine the root of the eukaryotic tree, researchers have based their phylogenies on different sets of genes, with conflicting results. Researchers have also tried a different approach, based on tracing the evolutionary history of genes that were transferred to eukaryotes from bacterial donors **(Figure 28.26)**. Results from this approach indicate that Excavata, SAR, and Archaeplastida share a more recent common ancestor than any of them does with Unikonta. This suggests that the root of the tree is located between the unikonts and all other eukaryotes, which implies that the unikonts were the first eukaryotic supergroup to diverge from all other eukaryotes. This idea remains controversial and will require more supporting evidence to be widely accepted.

Figure 28.26

Inquiry: What is the root of the eukaryotic tree?

Experiment

Responding to the difficulty in determining the root of the eukaryotic phylogenetic tree, Romain Derelle and colleagues used a new approach. They studied 39 nuclear genes that eukaryotes acquired by horizontal gene transfer from a single bacterial source: the alpha proteobacterium endosymbiont that gave rise to the mitochondrion. These 39 genes were known to be orthologous (see Figure 26.18). As a result, differences among these genes reflect the history of speciation events that have occurred since the eukaryotes began to diverge from their last common ancestor—making these genes well suited for inferring the root of the eukaryotic tree. Derelle and colleagues obtained the sequences of these genes for 67 species of eukaryotes representing a wide range of eukaryotic taxa. Those sequences were then used to infer evolutionary relationships among the four eukaryotic supergroups shown in Figure 28.5.

Results

A phylogenetic tree shows that a common ancestor of all eukaryotes branches to Unikonta and a clade which includes groups excavate, SAR, and Archaeplastida.

Conclusion

The results show that Excavata, SAR, and Archaeplastida form a clade, which supports the hypothesis that the root of the tree is located between the unikonts and all other eukaryotes.

Data from R. Derelle et al., Bacterial proteins pinpoint a single eukaryotic root, *Proceedings of the National Academy of Sciences USA*112:E693–699 (2015).

WHAT IF?*Other recent evidence indicates that the root of the eukaryotic tree may lie between a clade that includes unikonts and excavates and all other eukaryotes. Draw the tree suggested by this result.*

*For suggested answer, see***Appendix A*.

Amoebozoans

The **amoebozoan** clade includes many species of amoebas that have lobe- or tube-shaped pseudopodia rather than the threadlike pseudopodia found in rhizarians. Amoebozoans include tubulinids, slime molds, and entamoebas.

Tubulinids

Tubulinids constitute a large and varied group of amoebozoans that have lobe- or tube-shaped pseudopodia. These unicellular protists are ubiquitous in soil as well as freshwater and marine environments. Most are heterotrophs that actively seek and consume bacteria and other protists; one such tubulinid species, *Amoeba proteus*, is shown in Figure 28.5. Some tubulinids also feed on detritus (nonliving organic matter).

Slime Molds

Slime molds, or mycetozoans (from the Latin, meaning “fungus animals”), once were thought to be fungi because, like fungi, they produce fruiting bodies that aid in spore dispersal. However, DNA sequence analyses indicate that the resemblance between slime molds and fungi is a case of evolutionary convergence. Slime molds have diverged into two main branches, plasmodial slime molds and cellular slime molds. We’ll compare their characteristics and life cycles.

Plasmodial Slime Molds

Many plasmodial slime molds are brightly colored, often yellow or orange **(Figure 28.27)**. As they grow, they form a mass called a plasmodium, which can be many centimeters in diameter. (Don't confuse a slime mold's plasmodium with the genus *Plasmodium*, which includes the parasitic apicomplexan that causes malaria.) Despite its size, the plasmodium is not multicellular; it is a single mass of cytoplasm that is undivided by plasma membranes and that contains many nuclei. This "supercell" is the product of mitotic nuclear divisions that are not followed by cytokinesis. The plasmodium extends pseudopodia through moist soil, leaf mulch, or rotting logs, engulfing food particles by phagocytosis as it grows. If the habitat begins to dry up or there is no food left, the plasmodium stops growing and differentiates into fruiting bodies that function in sexual reproduction.

Figure 28.27

```
    A mature plasmodium, the feeding stage in the life cycle of
a plasmodial slime mold.
```

A micrograph of a plasmodium.

Cellular Slime Molds

The life cycle of the protists called cellular slime molds can prompt us to question what it means to be an individual organism. The feeding stage of these organisms consists of solitary cells that function individually, but when food is depleted, the cells form a slug-like aggregate that functions as a unit **(Figure 28.28)**. Unlike the feeding stage (plasmodium) of a plasmodial slime mold, these aggregated cells remain separated by their individual plasma membranes. Ultimately, the aggregated cells form an asexual fruiting body.

Figure 28.28

The life cycle of *Dictyostelium*, a cellular slime mold.

Life cycle diagram of Dictyostelium highlights haploid and diploid stages, sexual and asexual reproduction, as well as fertilization and meiosis.

The cycle is as follows.

1.In the feeding stage, solitary haploid amoebas (cells with lobe-like extensions) engulf bacteria; these solitary cells periodically divide by mitosis (asexual reproduction).

2.During sexual reproduction, two haploid amoebas fuse and form a diploid zygote.

3.The diploid zygote becomes a giant cell by consuming haploid amoebas. After developing a resistant wall, the giant cell undergoes meiosis followed by several mitotic divisions.

4.The wall ruptures, releasing new haploid amoebas.

5.When food is depleted, hundreds of amoebas congregate in response to a chemical attractant and form a haploid slug-like aggregate (photo shows slug-like form at scale 200 micrometers).

6.The haploid aggregate migrates for a while and then stops. Some of the cells dry up after forming a stalk that supports a haploid asexual fruiting body (Image shows long stalk with bulbous structure on the top at scale 600 micrometers).

7.Other cells crawl up the stalk of the haploid fruiting bodies and develop into haploid spores.

8.Haploid spores are released.

9.In favorable conditions, haploid amoebas emerge from the spore coats and feed, and the cycle can begin again at step 1.

VISUAL SKILLS *Suppose cells were removed from the slug-like aggregate shown in the photo. Use the information in the life cycle to infer whether these cells would be haploid or diploid. Explain.*

*For suggested answer, see**Appendix A.*

Dictyostelium discoideum, a cellular slime mold commonly found on forest floors, has become a model organism for studying the evolution of multicellularity. One line of research has focused on the slime mold's fruiting body stage. During this stage, the cells that form the stalk die as they dry out, while the spore cells at the top survive and have the potential to reproduce (see Figure 28.28). Scientists have found that mutations in a single gene can turn individual *Dictyostelium* cells into "cheaters" that never become part of the stalk. Because these mutants gain a strong reproductive advantage over noncheaters, why don't all *Dictyostelium* cells cheat?

Recent discoveries suggest an answer to this question. Cheating cells lack a specific surface protein, and noncheating cells can recognize this difference. Noncheaters preferentially aggregate with other noncheaters, thus depriving cheaters of the chance to exploit them. Such a recognition system may have been important in the evolution of other multicellular eukaryotes, such as animals and plants.

Entamoebas

Whereas most amoebozoans are free-living, those that belong to the genus *Entamoeba* are symbiotic parasites. They infect all classes of vertebrate animals as well as some invertebrates. Humans are host to at least six species of *Entamoeba*, but only one, *E. histolytica*, is known to be pathogenic. *E. histolytica* causes amoebic dysentery and is spread via contaminated drinking water, food, or eating utensils. Responsible for up to 110,000 deaths worldwide every year, the disease is the third-leading cause of death due to eukaryotic parasites, after malaria (see Figure 28.18) and schistosomiasis (see Figure 33.10).

Opisthokonts

Opisthokonts are an extremely diverse group of eukaryotes that includes animals, fungi, and several groups of protists. We will discuss the evolutionary history of fungi and animals in Chapters 31–34. Of the opisthokont protists, we will discuss the nucleariids in Chapter 31 because they are more closely related to fungi than they are to other protists. Similarly, we will discuss choanoflagellates in Chapter 32, since they are more closely related to animals than they are to other protists. The nucleariids and choanoflagellates illustrate why scientists have abandoned the former kingdom Protista: A monophyletic group that includes these single-celled eukaryotes would also have to include the multicellular animals and fungi that are closely related to them.

Concept Check 28.5

- Contrast the pseudopodia of amoebozoans and forams.
- In what sense is "fungus animal" a fitting description of a slime mold? In what sense is it not fitting?
- **MAKE CONNECTIONS** Alpha proteobacteria were used as the outgroup to estimate the tree shown in Figure 28.26. Explain why these bacteria were selected as the outgroup rather than other bacterial or archaeal lineages. (See Figure 26.12 and associated text in Concept 26.3.)

*For suggested answers, see*Appendix A.**

Concept 28.6: Protists play key roles in ecological communities

Most protists are aquatic, and they are found almost anywhere there is water, including moist terrestrial habitats such as damp soil and leaf litter. In oceans, ponds, and lakes, many protists are bottom-dwellers that attach to rocks and other substrates or creep through the sand and silt. As we've seen, other protists are important constituents of plankton. We'll focus here on two key roles that protists play in the varied habitats in which they live: that of symbiont and that of producer.

Symbiotic Protists

Many protists form symbiotic associations with other species. For example, photosynthetic dinoflagellates are food-providing symbiotic partners of the animals (coral polyps) that build coral reefs. Coral reefs are highly diverse ecological communities. That diversity ultimately depends on corals—and on the mutualistic

protists that nourish them. Corals support reef diversity by providing food to some species and habitat to many others.

Another example is the wood-digesting protists that inhabit the gut of many termite species **(Figure 28.29)**. Unaided, termites cannot digest wood, and they rely on protistan or prokaryotic symbionts to do so. Termites cause over $3.5 billion in damage annually to wooden homes in the United States.

Figure 28.29

```
A symbiotic protist.
```

Wood-digesting protist appears “hairy” and dome-shaped with a smooth “cap” in this image. Scale: 10 micrometers.

This organism is a hypermastigote, a member of a group of parabasalids that live in the gut of termites and certain cockroaches and enable the hosts to digest wood (SEM).

Symbiotic protists also include parasites that have compromised the economies of entire countries. Consider the malaria-causing protist *Plasmodium*: Income levels in countries hard hit by malaria are 33% lower than in similar countries free of the disease. Protists can have devastating effects on other species too. Massive fish kills have been attributed to *Pfiesteria shumwayae* (see Figure 28.17), a dinoflagellate parasite that attaches to its victims and eats their skin. Among species that parasitize plants, the stramenopile *Phytophthora ramorum* (an oomycete) has emerged as a major new forest pathogen. This species causes sudden oak death (SOD), a disease that has killed millions of oaks and other trees in the United States and Great Britain (**Figure 28.30**; also see Concept 54.5).

Figure 28.30

```
Sudden oak death.
```

Aerial photograph of Monterey County, California landscape, showing dead oak trees.

Many dead oak trees are visible in this Monterey County, California landscape. Infected trees lose their ability to adjust to cycles of wet and dry weather.

Photosynthetic Protists

Many protists are important **producers**, organisms that use energy from light (or in some prokaryotes, inorganic chemicals) to convert CO_2 to organic compounds. Producers form the base of ecological food webs.

In aquatic communities, the main producers are photosynthetic protists and prokaryotes **(Figure 28.31)**. All other organisms in the community are **consumers** that depend on producers for food, either directly (by eating them) or indirectly (by eating an organism that ate a producer). Scientists estimate that roughly 30% of the world's photosynthesis is performed by diatoms, dinoflagellates, multicellular algae, and other aquatic protists. Photosynthetic prokaryotes contribute another 20%, and plants are responsible for the remaining 50%.

Figure 28.31

```
Protists: key producers in aquatic communities.
```

Herbivorous plankton are food for carnivorous plankton and other consumers (fish, orcas, penguins, invertebrates). Prokaryotic and protistan producers are food for herbivorous plankton. Carnivorous plankton are food for other consumers.

Arrows in this simplified food web lead from food sources to the organisms that eat them.

Because producers form the foundation of food webs, factors that affect producers can dramatically affect their entire community. In aquatic environments, photosynthetic protists are often held in check by low concentrations of nitrogen, phosphorus, or iron. Various human actions can increase the concentrations of these elements in aquatic communities. For example, when fertilizer is applied to a field, some of the fertilizer may be washed by rainfall into a river that drains into a lake or ocean. When people add nutrients to aquatic communities in this or other ways, the abundance of photosynthetic protists can increase spectacularly. Such increases can have major ecological consequences, including the formation of large "dead zones" in marine ecosystems (see Figure 56.23).

A pressing question is how global warming will affect photosynthetic protists and other producers. As shown in **Figure 28.32**, the growth and biomass of photosynthetic protists and prokaryotes have declined in many ocean regions as sea surface temperatures have increased. By what mechanism do rising sea surface temperatures reduce the growth of marine producers? One hypothesis relates to the rise or upwelling of cold, nutrient-rich waters from below. Many marine producers rely on nutrients brought to the surface in this way. However, rising sea surface temperatures can cause the formation of a layer of light, warm water that acts as a barrier to nutrient upwelling—thus reducing the growth of marine producers. If sustained, the changes shown in Figure 28.32 would likely have far-reaching effects on marine ecosystems, fishery yields, and the global carbon cycle (see Figure 55.14). Global warming can also affect producers on land, but there the base of food webs is occupied not by protists but by plants, which we will discuss in Chapters 29 and 30.

Figure 28.32

```
Effects of climate change on marine producers.
```

A map and bar chart illustrate effects of climate change.

Parts are as follows.

Part a. Researchers studied 10 ocean regions, identified with letters on the map (see (b) for the corresponding names). SSTs have increased since 1950 in most areas of these regions. Map of the world shows changing sea surface temperatures (SSTs) in 10 ocean regions. Ranges for Sea Surface Temperature Change in degrees Celsius are negative 2.50 to negative 1.25; negative 1.25 to 0.00; 0.00 to 1.25; 1.25 to 2.50. Sea surface temperatures have risen in the following oceans: South Indian, North Indian, Equatorial Pacific, Arctic, South Pacific, North Atlantic, Equatorial Atlantic, and South Atlantic. Ocean regions that show small areas of the highest warming (+ 2.5 degrees Celsius) are in the South Indian Ocean, the South Atlantic off the east coast of South America, the North Atlantic off the east coast of Canada, and various regions in the Arctic Ocean. There are pockets of temperature increase in the North Pacific, Equatorial Pacific, Atlantic, and Southern Ocean regions.

Part b. The concentration of chlorophyll, an index for the biomass and growth of marine producers, has decreased over the same time period in most ocean regions. Bar chart of ocean region (same as those in part a) by chlorophyll change in milligrams per meter squared per year. Chlorophyll concentration change ranges from -0.02 to 0.02 with increments every 0.01 In all ocean regions except the North and South Indian, chlorophyll concentrations have declined.

Concept Check 28.6

- Justify the claim that photosynthetic protists are among the biosphere's most important organisms.
- Describe three symbioses that include protists.
- **WHAT IF?** High water temperatures and pollution can cause corals to expel their dinoflagellate symbionts. How might such "coral bleaching" affect corals and other species?
- **MAKE CONNECTIONS** The bacterium *Wolbachia* is a symbiont that lives in mosquito cells and spreads rapidly through mosquito populations. *Wolbachia* can make mosquitoes resistant to infection by *Plasmodium*; researchers are seeking a strain that confers resistance and does not harm mosquitoes. Compare evolutionary changes that could occur if malaria control is attempted using such a *Wolbachia* strain versus using insecticides to kill mosquitoes. (Review Figure 28.18 and Concept 23.4.)

*For suggested answers, see*Appendix A.**